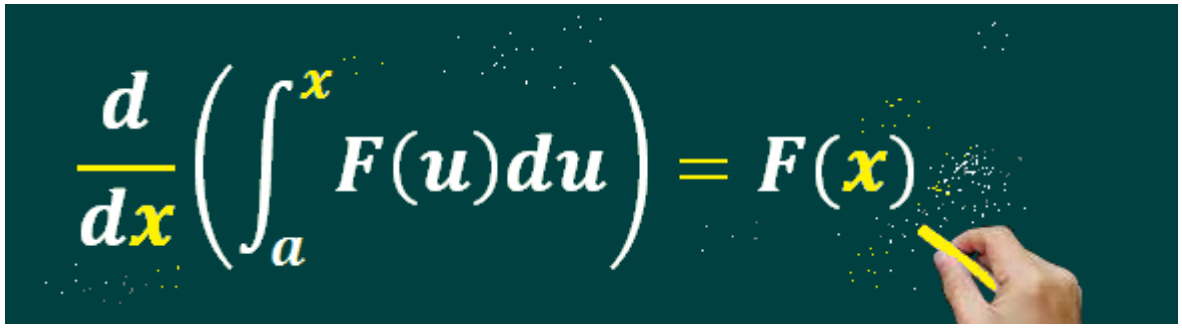
Refresher Course in Mathematics 2023

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Session Two

Lessons in Elementary Functions and Single Variable Calculus
Chapter 1, Sections 1.3 and 1.4



Equations

An equation is a mathematical equality relation containing one or more unknown quantities.

E.g. $2x+1=0$ $y^3-y=0$ $a^2=b^2+c^2$

A solution of an equation is given by the values of the unknown quantities such that the LHS of the equation is equal to the RHS



$x=-\frac{1}{2}$ is a (the only) solution of $2x+1=0$.

$y^3-y=0 \Rightarrow y(y^2-1)=0 \Rightarrow y=0$ or $y^2-1=0$

$\Rightarrow y=0, y=-1, y=1$

$a^2=b^2+c^2$

Let $b=3$, $c=4$

$\Rightarrow a^2=25 \Rightarrow a=5$ or $a=-5$

Some solutions are $(a,b,c)=(5,3,4)$ and $(a,b,c)=(-5,3,4)$

Quadratic equations

$$ax^2+bx+c=0$$

x is the unknown

a, b and c are real numbers (independent of x)

$a \neq 0$

E.g.

$x^2-5=0 \Rightarrow x^2=5 \Rightarrow x=5^{1/2}$ or $x=-5^{1/2}$

$y^2+1=0 \Rightarrow y^2=-1$

No solution as far as real numbers are concerned

$x^2-2x-15=0$

$\Rightarrow \underbrace{(x-1)^2}_{x^2-2x+1}-15-1=0$

Complete the square



$\Rightarrow (x-1)^2-16=0$

$\Rightarrow (x-1)^2=16$

$\Rightarrow x-1=4$ or $x-1=-4$

$\Rightarrow x=5$ or $x=-3$

Quadratic equations

$$ax^2+bx+c=0 \qquad a\neq 0$$

$$\Rightarrow \quad x^2+\frac{b}{a}x+\frac{c}{a}=0$$

$$\Rightarrow \left(x+\frac{b}{2a}\right)^2+\frac{c}{a}-\frac{b^2}{4a^2}=0$$



Complete the square

$$x^2+\frac{b}{a}x+\frac{b^2}{4a^2}$$

$$\Rightarrow \left(x+\frac{b}{2a}\right)^2=\frac{b^2}{4a^2}-\frac{c}{a}=\frac{b^2-4ac}{4a^2}$$

$$\Rightarrow x+\frac{b}{2a}=\pm\sqrt{\frac{b^2-4ac}{(2a)^2}}=\pm\frac{\sqrt{b^2-4ac}}{2a}$$

$$\Rightarrow \boxed{x=-\frac{b\pm\sqrt{b^2-4ac}}{2a}}$$

$b^2-4ac=0$ ⇒ one real soln

$b^2-4ac>0$ ⇒ two real solns

$b^2-4ac<0$ No real soln

Quadratic equations

If the quadratic equation $ax^2+bx+c=0$ $(a\neq 0)$ can be rewritten as

$$a(x-\alpha)(x-\beta)=0$$

factorisation of ax^2+bx+c

then the quadratic equation has solutions given by $x=\alpha$ and $x=\beta$.

E.g. $2x^2-5x-3=0$

$\Rightarrow (2x+1)(x-3)=0$

$\Rightarrow 2\left(x+\frac{1}{2}\right)(x-3)=0$

$\Rightarrow x=-\frac{1}{2}$ and $x=3$ are sol'ns



Polynomial equations

N^{th} order polynomial equation

A polynomial equation of order N is an equation of the form

N is a positive integer

$$a_N x^N + a_{N-1} x^{N-1} + \ldots + a_2 x^2 + a_1 x + a_0 = 0$$



x is unknown

$a_0, a_1, a_2, \ldots, a_{N-1}$ and a_N are real numbers independent of x. $a_N \neq 0$

e.g.

1st order polynomial eqn (linear equation) $2x+1=0$

2nd order polynomial eqn (quadratic eqn) $x^2 - 9 = 0$

3rd order polynomial eqn (cubic eqn) $3x^3 + x^2 + x + 2 = 0$

4th order polynomial eqn (quartic eqn) $x^4 + 2x = 0$

5th order polynomial eqn (quintic eqn) $-2x^5 - x^3 - x = 1$

6th order polynomial eqn (sextic eqn) $x^6 = 64$

Polynomial equations

If $x = c$ is a solution of the N-th order polynomial eqn $a_N x^N + a_{N-1} x^{N-1} + \ldots + a_1 x + a_0 = 0$ for $N \geq 2$, we can rewrite the polynomial eqn as

$$a_N (x-c)\left(x^{N-1} + b_{N-2} x^{N-2} + b_{N-3} x^{N-3} + \ldots + b_1 x + b_0\right) = 0$$

Note $a_N \neq 0$

The remaining solutions of $a_N x^N + a_{N-1} x^{N-1} + \ldots + a_1 x + a_0 = 0$, if there is any, can be obtained by sdving

$$x^{N-1} + b_{N-2} x^{N-2} + b_{N-3} x^{N-3} + b_1 x + b_0 = 0.$$

$(N-1)^{th}$ order polynomial eqn



Polynomial equations

E.g. $2x^3 + 2x^2 - 3x - 1 = 0$ ..... (1)



$x = 1$ is a solution of (1).

We can rewrite (1) as

$$2(x-1)(x^2 + ax + b) = 0 \quad \text{..... (2)}$$

If we expand the LHS of (2), we obtain

$$2x^3 + (2a-2)x^2 + (2b-2a)x - 2b = 0 \quad \text{..... (3)}$$

Compare (1) and (3):

$2a - 2 = 2$, $2b - 2a = -3$, $-2b = -1$

$\Rightarrow b = \frac{1}{2}$ $a = 2$

The remaining solutions can be obtained by solving

$$x^2 + 2x + \frac{1}{2} = 0 \quad \Rightarrow x = \frac{-1 \pm \sqrt{4-2}}{2}$$

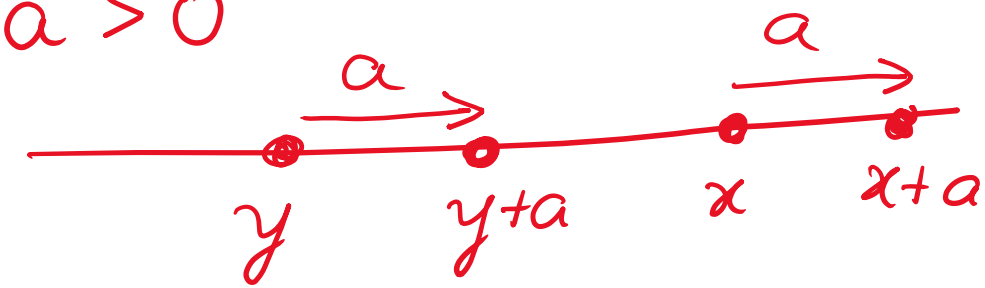
Properties of inequalities

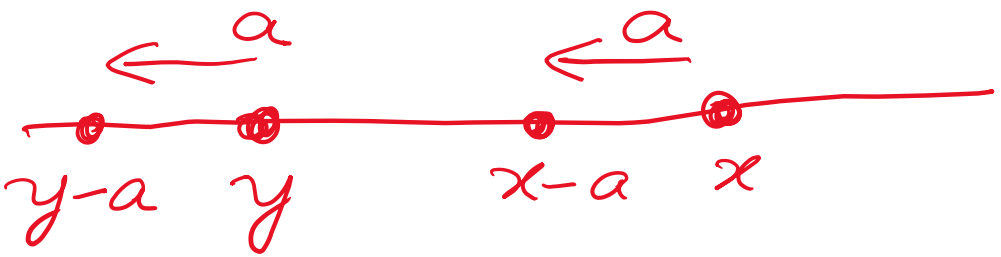
Suppose $x > y$.



$\Rightarrow x+a > y+a$ and $x-a > y-a$

Assume $a > 0$





Likewise,

if $x < y$ then $x+a < y+a$

and $x-a < y-a$.



Properties of inequalities



eg. $3 > 2 \Rightarrow 10(3) > 10(2)$,

$30 > 20$.

If $x > y$ then $ax > ay$ for any positive real no. a.

$-10(3) < -10(2)$, $\quad -30 < -20$

If $x > y$ then $ax < ay$ for any negative real no. a

$x < y \Rightarrow ax < ay$ for any positive real no. a

$x < y \Rightarrow ax > ay$ for any negative real no. a.

Properties of inequalities

If $x > y$ then $\frac{1}{a}x > \frac{1}{a}y$

for any positive real no.

and $\frac{1}{a}x < \frac{1}{a}y$

for any negative real no.

If $x < y$ then

$\frac{1}{a}x < \frac{1}{a}y$ for any real positive no

and

$\frac{1}{a}x > \frac{1}{a}y$ for any real negative no.



Solving inequality relations

Find all the values of x such that the inequality is true. Rewrite, if needed, the inequality such that one side is given by 0

Eg. Solve $2x^3+2x^2-3x<1$

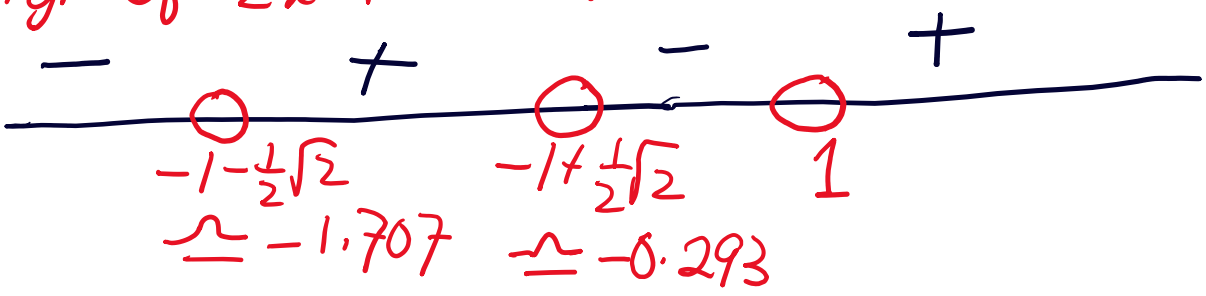
$\Rightarrow 2x^3+2x^2-3x-1<0$

To construct the sign diagram of $2x^3+2x^2-3x-1$, first find all values of x where $2x^3+2x-3x-1$ can change signs.

We solve $2x^3+2x^2-3x-1=0$ because $2x^3+2x^2-3x-1$ can possibly change at where $2x^3+2x^2-3x-1$ is 0.

Solutions of $2x^3+2x^2-3x-1=0$ are $x=1$, $x=-1-\frac{1}{2}\sqrt{2}$, $x=-1+\frac{1}{2}\sqrt{2}$

Sign of $2x^3+2x^2-3x-1$



$\Rightarrow x \in (-\infty, -1-\frac{1}{2}\sqrt{2}) \cup (-1+\frac{1}{2}\sqrt{2}, 1)$



Solving inequality relations

E.g. Solve $\frac{x}{(x+1)^2} \geq \frac{1}{x-3}$

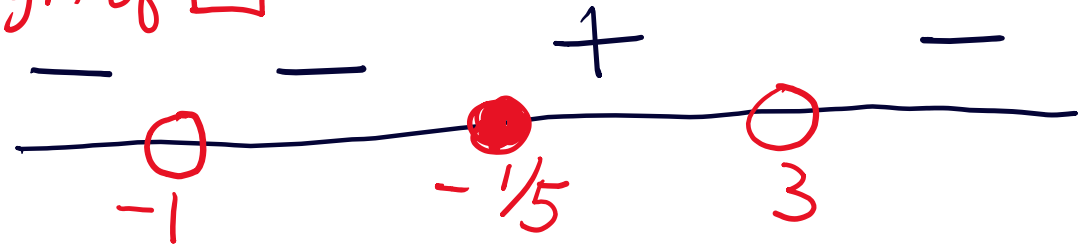
$\Rightarrow \quad \frac{x}{(x+1)^2} - \frac{1}{(x-3)} \geq 0 \quad \Rightarrow \boxed{\frac{-5x-1}{(x+1)^2(x-3)} \geq 0}$

$\square$ may possibly change signs at where $\square$ is zero or undefined.

$\square = 0$ at $x = -\frac{1}{5}$

$\square$ is undefined at $x = -1$ and $x = 3$

Sign of $\square$



$\Rightarrow \quad x \in \left[-\frac{1}{5}, 3\right)$



Selected problems in Section 1.5

Problem 1(b) If x and y are rational numbers, show that $x-y$ is a rational number.

x is rational $\Rightarrow$ we can write $x = \frac{m_1}{n_1}$ where m_1 and n_1 are integers

y is rational $\Rightarrow$ we can write $y = \frac{m_2}{n_2}$ where m_2 and n_2 are integers.

$$x - y = \frac{m_1}{n_1} - \frac{m_2}{n_2} = \frac{n_2 m_1 - m_2 n_1}{n_1 n_2}$$

$n_2 m_1 - m_2 n_1$ → INTEGER

$n_1 n_2$ → INTEGER

$\Rightarrow x - y$ is rational



Selected problems in Section 1.5

Problem 2(a) Prove or disprove:

"If x is rational and y is irrational then $x+y$ is irrational."

THIS IS TRUE!!

Let $z = x+y$.

Assume z is rational.

We are told x is rational and y is irrational.

From $z = x+y$, we can write:

$$y = x - z$$

(x: RATIONAL; z: ASSUMED RATIONAL)

If z is assumed rational, from Problem 1(b), we have to conclude y is rational.

But y is irrational (as given in problem)

$\Rightarrow z$ cannot be rational. $\Rightarrow z$ is irrational



Selected problems in Section 1.5

NOT TRUE

Problem 2(d) Prove or disprove:

"If x and y are irrational then xy is irrational."

Let $x = \pi$ (irrational)

$y = \frac{1}{\pi}$ (irrational)

$xy = \pi \times \frac{1}{\pi} = 1$ (rational)

Proof by counter-example



Selected problems in Section 1.5

Problem 3 (a) Express $0.\overline{76}$ in the form $\frac{m}{n}$, where m and n are integers

$$0.\overline{76} = 0.76\,76\,76\,76\ldots$$

$$= \frac{76}{10^2} + \frac{76}{10^4} + \frac{76}{10^6} + \frac{76}{10^8} + \cdots\cdots$$

$$= \frac{76}{100}\left(1 + \frac{1}{10^2} + \frac{1}{10^4} + \frac{1}{10^6} + \cdots\right)$$

$$= \frac{76}{100} \times \frac{1}{1-\frac{1}{100}} = \frac{76}{100} \times \frac{100}{99}$$

$$= \frac{76}{99}$$



Selected problems in Section 1.5

Problem 10(d) Solve $|x-1| \leq |x-7|$.

If we assume $x \neq 7$, we can rewrite the above inequality as: $\dfrac{|x-1|}{|x-7|} \leq 1$

Is $x=7$ a solution? We can check.

$|x-1| = 6$ if $x=7$

$|x-7| = 0$ if $x=7$

$6 \geq 0$. So $|x-1| \leq |x-7|$ is not true

$x=7$ is not a solution.

We use $\dfrac{|a|}{|b|} = \left|\dfrac{a}{b}\right|$ to rewrite $\dfrac{|x-1|}{|x-7|} \leq 1$ as:

$$\left|\frac{x-1}{x-7}\right| \leq 1.$$



Selected problems in Section 1.5



$|x| \le a \Rightarrow -a \le x \le a$.

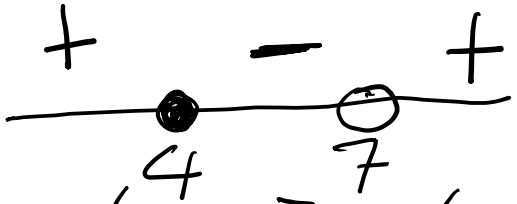
$$\left|\frac{x-1}{x-7}\right| \le 1 \Rightarrow -1 \le \frac{x-1}{x-7} \le 1$$

So we have to solve:

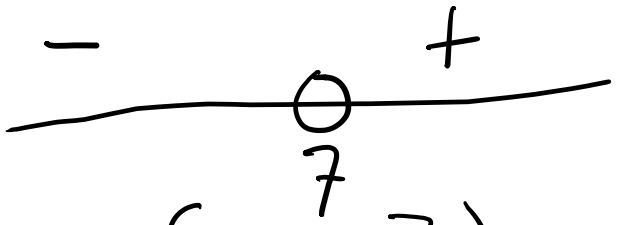
$$\frac{x-1}{x-7} \ge -1 \quad \text{AND} \quad \frac{x-1}{x-7} \le 1$$

$$\Rightarrow \frac{x-1}{x-7} + 1 \ge 0 \quad \text{AND} \quad \frac{x-1}{x-7} - 1 \le 0$$

$$\Rightarrow \frac{2x-8}{x-7} \ge 0 \quad \text{AND} \quad \frac{8}{x-7} \le 0$$



AND



$x \in (-\infty, 4] \cup (7, \infty)$ AND $x \in (-\infty, 7)$

$\Rightarrow x \in (-\infty, 4]$

Any question?



A yawning long-tailed macaque at Singapore Chinese Gardens
Image courtesy of Robert Ang of “I walked, I spotted, I captured”